

# Calm Months, Crash Months: Volatility-Managed Factors in Real Time and After Publication

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## Abstract

Volatility-managed factors look strong in spanning regressions, but those regressions describe a portfolio no investor could have formed in advance. We rebuild the Fama-French five factors and momentum so that each month's exposure uses only prior data, is capped at 1.5 times, and pays trading costs, and we compare Sharpe ratios with a studentized block bootstrap. From July 1973 to August 2026, only momentum improves significantly, from 0.43 to 0.90. The gain is concentrated in the years before the published samples end: from 2017 onward, managed momentum matches the original. No real-time combination of managed and original factor beats its original once it pays the same costs. An exact decomposition of the Sharpe ratio into a return channel and a risk channel shows why: before 2017 both channels lifted momentum; afterwards trading costs absorbed the timing return and the risk channel had little left to cut.

**Keywords:** volatility-managed portfolios, factor timing, momentum, out-of-sample performance, post-publication decay

**JEL:** G11, G12, G14

## 1 Introduction

Moreira and Muir (2017) show that scaling a factor's exposure by the inverse of its recent variance raises the Sharpe ratio of many equity factors. Their evidence is a spanning regression of the managed factor on the original. A positive intercept means that some mix of the two beats the original, but the mix uses full-sample moments no investor had in advance. Cederburg et al. (2020) make the point directly: across 103 strategies ending in December 2016, real-time combinations of managed and original portfolios usually fail to beat the original. Liu et al. (2019) show that the market-timing version relies on a scale constant fitted with future data, and Barroso and Detzel (2021) show that transaction costs erase most of the gains, with the market and momentum partly robust. Momentum is the most consistent survivor, in line with the crash evidence of Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016).

This letter asks two questions. First, which of the six most widely used equity factors improve once every position uses only past data, leverage is capped and costs are paid? Second, what

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happened after the published samples ended? Moreira and Muir’s data end in 2015 and Cederburg et al.’s in 2016, so January 2017 to August 2026 is out of sample for both. McLean and Pontiff (2016) find that anomaly returns are 58% lower after publication, which makes this decade a natural test.

The answers are narrow. Only momentum’s managed version beats the original over the full sample. That gain is concentrated before 2017; afterwards managed and original momentum have nearly the same Sharpe ratio, although the decade is too short to prove the gain has gone. Profitability, whose managed version earns a significant spanning alpha in Cederburg et al. (2020), shows no significant real-time gain. An exact decomposition of the Sharpe ratio explains the pattern: volatility management works through a return channel, the covariance between exposure and returns net of costs, and a risk channel, the volatility it removes relative to a constant exposure.

## 2 Data and method

We use the market, size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum factors of Fama and French (2015) and Carhart (1997) from the Kenneth R. French Data Library, July 1963 to August 2026. Monthly returns  $f_m$  come from the monthly files. Daily returns  $r_d$  serve only to measure risk. Realized variance follows Moreira and Muir (2017):

$$\text{RV}_m = \sum_{d \in m} (r_d - \bar{r}_m)^2, \quad (1)$$

where  $\bar{r}_m$  is the mean daily return in month  $m$ . French’s daily momentum factor is re-formed daily, so for momentum  $\text{RV}_m$  is a proxy for the variance of the monthly-formed factor. We use the current vintage of the data; French revises the history as CRSP and Compustat are updated, so positions use only past returns but not the returns as first published.

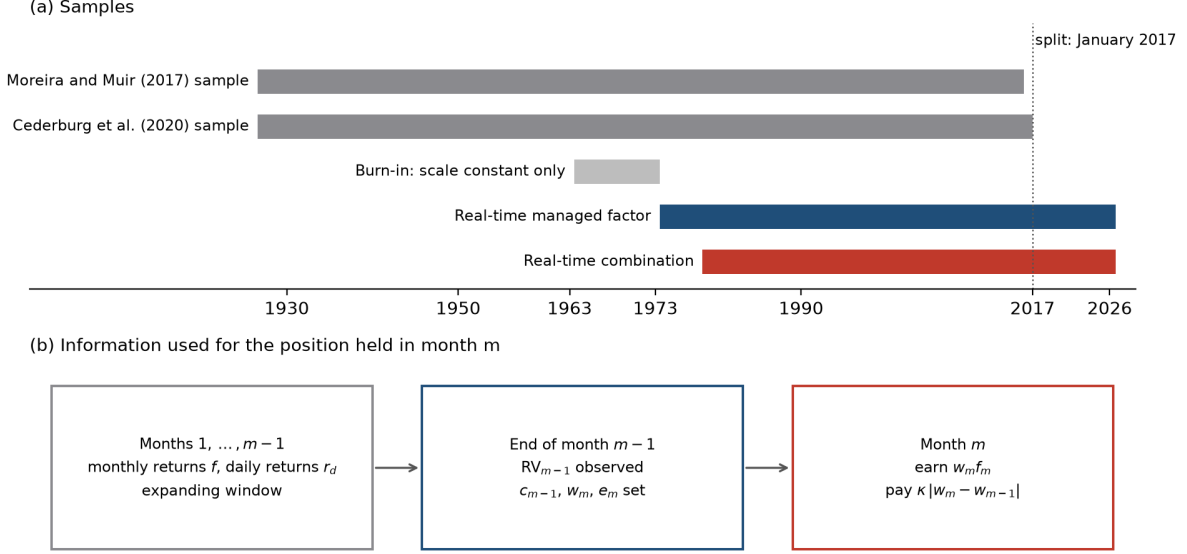
**Managed factor.** The exposure held in month  $m$  is set at the end of month  $m - 1$  (Figure 1):

$$w_m = \min\left(\frac{c_{m-1}}{\text{RV}_{m-1}}, L\right), \quad c_{m-1} = \frac{\text{sd}(f_1, \dots, f_{m-1})}{\text{sd}(f_s/\text{RV}_{s-1} : s \leq m-1)}. \quad (2)$$

The scale  $c_{m-1}$  is the expanding-window version of Moreira and Muir’s full-sample constant: it makes the uncapped managed factor as volatile as the original over the data seen so far. The first 120 months are a burn-in, so positions start in July 1973 (638 months). The baseline cap is  $L = 1.5$ , the higher of the two caps (1 and 1.5) in Moreira and Muir’s Table IV. Changing exposure costs  $\kappa = 14$  basis points per unit, the highest of their cost levels, so the managed return is

$$f_m^\sigma = w_m f_m - \kappa |w_m - w_{m-1}|. \quad (3)$$

**Spanning regression.**  $f_m^\sigma = \alpha + \beta f_m + \varepsilon_m$ , with Newey-West  $t$ -statistics (six lags). Because  $f^\sigma$  is net of costs,  $\alpha$  is the return a static position in the original could not replicate, after costs.



**Figure 1:** Study design. Panel (a): sample windows. The published samples start in the late 1920s for the market, size, value and momentum factors and in 1963 for profitability and investment; Moreira and Muir’s end in 2015 and Cederburg et al.’s in December 2016. Our first ten years only feed the scale constant. Panel (b): the position held in month  $m$  uses only information available at the end of month  $m - 1$  (equations (2) to (4)).

**Real-time combination.** Following Cederburg et al. (2020), from month 60 of the real-time sample the investor forms the mean-variance mix of the original and managed factor using the expanding window  $s < m$ :

$$\begin{pmatrix} a_m \\ b_m \end{pmatrix} = \hat{\Sigma}_{m-1}^{-1} \hat{\mu}_{m-1}, \quad e_m = \text{clip}(k_m (a_m + b_m w_m), -L, L), \quad (4)$$

where  $\hat{\mu}$  and  $\hat{\Sigma}$  are the mean and covariance of  $(f_s, f_s^\sigma)$  and  $k_m$  scales the mix to the original factor’s volatility. Since  $f^\sigma = w f$  before costs, the mix is a single exposure  $e_m$  to the original factor, capped like the managed factor and charged the same cost:  $f_m^c = e_m f_m - \kappa |e_m - e_{m-1}|$ .

**Inference.** For two return series with sample moments  $\hat{v} = (\hat{\mu}_1, \hat{\mu}_2, \hat{\gamma}_1, \hat{\gamma}_2)$ , where  $\hat{\gamma}_i$  is the mean of squared returns, the Sharpe difference is

$$\hat{\Delta} = g(\hat{v}) = \frac{\hat{\mu}_1}{\sqrt{\hat{\gamma}_1 - \hat{\mu}_1^2}} - \frac{\hat{\mu}_2}{\sqrt{\hat{\gamma}_2 - \hat{\mu}_2^2}}, \quad \text{se}(\hat{\Delta}) = \sqrt{\nabla g(\hat{v})^\top \hat{\Psi} \nabla g(\hat{v}) / T}, \quad (5)$$

with  $\hat{\Psi}$  a Bartlett long-run covariance of the four moment series, with lag length equal to the bootstrap block length. We test  $\Delta = 0$  with the studentized circular block bootstrap of Ledoit and Wolf (2008): 4,999 resamples of 12-month blocks, each with its own block-based standard error  $\text{se}^*$ . The  $p$ -value is  $(1 + \#\{|\hat{\Delta}^* - \hat{\Delta}| / \text{se}^* \geq |\hat{\Delta}| / \text{se}\}) / 5000$ , and the 95% interval is  $\hat{\Delta} \pm q_{0.95}^* \text{se}$ , where  $q_{0.95}^*$  is the 95th percentile of the resampled statistic. Within each family of six factors,  $p$ -values are

adjusted with the step-down procedure of Holm (1979): with ordered  $p_{(1)} \leq \dots \leq p_{(6)}$ , the adjusted value is  $\tilde{p}_{(j)} = \max_{i \leq j} \min\{1, (7 - i) p_{(i)}\}$ . A test that the Sharpe gain differs across two periods treats the periods as independent and combines their bootstrap draws in the same studentized way.

**Why timing can help or hurt.** For any exposure series, the mean managed return decomposes exactly as

$$\overline{f^\sigma} = \bar{w} \bar{f} + \widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|, \quad (6)$$

with the covariance taken over the same months. The first term is what a constant exposure  $\bar{w}$  would earn, and a constant exposure has the original factor's Sharpe ratio. Dividing the managed Sharpe ratio by the original's therefore gives, whenever  $\bar{f} > 0$ ,

$$\frac{\text{SR}(f^\sigma)}{\text{SR}(f)} = \underbrace{\left(1 + \frac{\widehat{\text{cov}}(w_m, f_m) - \kappa |\overline{\Delta w}|}{\bar{w} \bar{f}}\right)}_{M: \text{ return channel}} \times \underbrace{\frac{\bar{w} \text{sd}(f)}{\text{sd}(f^\sigma)}}_{R: \text{ risk channel}}. \quad (7)$$

$M$  measures the return that timing adds, net of costs, relative to a constant exposure of the same average size;  $R$  measures how much volatility timing removes relative to that exposure. Moreira and Muir (2017) motivate  $w \propto 1/\sigma^2$  with the mean-variance weight  $w_m^* = \mu_m/(\gamma\sigma_m^2)$ . If the expected return  $\mu_m$  does not vary with variance, the covariance is zero in expectation and the whole gain comes through  $R$ . If  $\mu_m$  is lower in turbulent months, as in the momentum crashes of Daniel and Moskowitz (2016), the covariance adds a second gain through  $M$ . If  $\mu_m$  is higher in turbulent months,  $M$  falls below one and the rule scales down exactly when the factor pays.

### 3 Results

#### 3.1 The full real-time sample

| Factor              | Managed factor, net |             |              | Real-time combination, net |             |            | Largest loss, equal volatility |         |
|---------------------|---------------------|-------------|--------------|----------------------------|-------------|------------|--------------------------------|---------|
|                     | Original            | Managed     | $p$ (Holm)   | Original                   | Combination | $p$ (Holm) | Original                       | Managed |
| Market              | 0.51                | 0.51        | 1.000        | 0.57                       | 0.47        | 0.540      | −68                            | −39     |
| Size (SMB)          | 0.21                | 0.12        | 1.000        | 0.10                       | −0.02       | 0.151      | −77                            | −92     |
| Value (HML)         | 0.32                | 0.23        | 1.000        | 0.24                       | 0.17        | 0.064      | −79                            | −76     |
| Profitability (RMW) | 0.38                | 0.47        | 1.000        | 0.47                       | 0.34        | 0.540      | −51                            | −37     |
| Investment (CMA)    | 0.45                | 0.34        | 1.000        | 0.37                       | 0.31        | 0.358      | −31                            | −53     |
| Momentum            | 0.43                | <b>0.90</b> | <b>0.029</b> | 0.42                       | 0.74        | 0.358      | −76                            | −23     |

**Table 1:** Annualized Sharpe ratios, July 1973 to August 2026 (638 months), exposure capped at 1.5, costs of 14 basis points per unit of exposure change. The combination starts in July 1978; its first column is the original factor over the same months.  $p$ -values are Ledoit-Wolf studentized block-bootstrap tests of equal Sharpe ratios, Holm-adjusted across the six factors; bold marks adjusted  $p < 0.05$ . Largest loss is the maximum peak-to-trough decline of summed monthly returns, in percentage points, with each series scaled to the original factor's volatility.

Table 1 reports the baseline. Managed momentum raises the Sharpe ratio from 0.43 to 0.90 after costs (95% interval for the gain 0.16 to 0.78;  $p = 0.005$ , Holm-adjusted 0.029). At equal volatility,

its largest loss falls from 76 to 23 percentage points. The gain would survive costs of up to 121 basis points per unit of exposure change, about 9 times the assumed level.

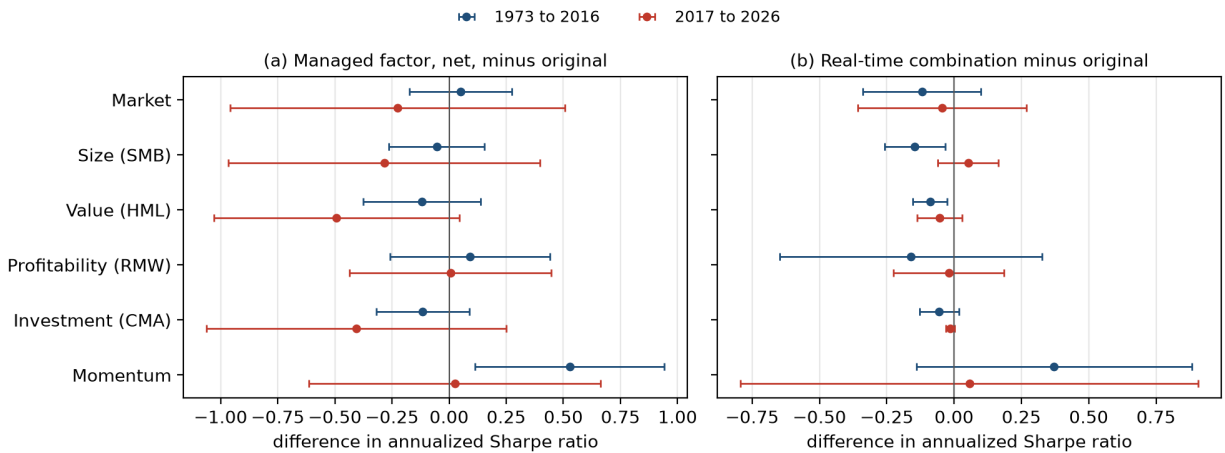
No other factor improves significantly. Profitability is the instructive case: its managed version earns a spanning alpha significant at 1% in Cederburg et al. (2020), but in real time its net spanning alpha is 0.9% a year ( $t = 1.9$ ) and its Sharpe ratio moves from 0.38 to 0.47 ( $p = 0.537$ , unadjusted). Cederburg et al. (2020) trace this gap between in-sample alpha and real-time gains to unstable spanning regressions.

The real-time combination does not beat any original factor once it pays costs and respects the cap. Even for momentum it rises only from 0.42 to 0.74 ( $p = 0.097$ , unadjusted). Removing the cap changes little: the Sharpe ratio rises to 0.77 ( $p = 0.062$ ). Removing costs as well, as in Cederburg et al. (2020), lifts it to 0.85 ( $p = 0.030$ ), consistent with their finding that momentum’s combination adds value before frictions. Costs, more than the cap, take away its significance.

### 3.2 After the published samples

| Factor              | 1973 to 2016 (522 months) |         |            |                      | 2017 to 2026 (116 months) |         |            |                      |
|---------------------|---------------------------|---------|------------|----------------------|---------------------------|---------|------------|----------------------|
|                     | Original                  | Managed | $p$ (Holm) | $\alpha$ , % ( $t$ ) | Original                  | Managed | $p$ (Holm) | $\alpha$ , % ( $t$ ) |
| Market              | 0.44                      | 0.49    | 1.000      | 1.3 (1.50)           | 0.81                      | 0.59    | 1.000      | 0.1 (0.04)           |
| Size (SMB)          | 0.31                      | 0.25    | 1.000      | 0.0 (0.08)           | -0.25                     | -0.53   | 1.000      | -1.5 (-1.68)         |
| Value (HML)         | 0.45                      | 0.33    | 1.000      | 0.0 (0.04)           | -0.12                     | -0.61   | 0.388      | -1.6 (-2.74)         |
| Profitability (RMW) | 0.42                      | 0.51    | 1.000      | 1.0 (1.79)           | 0.26                      | 0.27    | 1.000      | 0.2 (0.42)           |
| Investment (CMA)    | 0.62                      | 0.51    | 1.000      | 0.2 (0.40)           | -0.11                     | -0.52   | 0.924      | -1.7 (-2.26)         |
| Momentum            | 0.48                      | 1.00    | 0.116      | 5.2 (4.91)           | 0.21                      | 0.23    | 1.000      | 0.3 (0.44)           |

**Table 2:** Real-time managed factors (cap 1.5, net of costs) before and after the published samples end. Sharpe ratios are annualized;  $p$ -values are Holm-adjusted within each period;  $\alpha$  is the annualized net spanning alpha with its Newey-West  $t$ -statistic.



**Figure 2:** Sharpe ratio gain over the original factor, with 95% Ledoit-Wolf studentized bootstrap intervals, before and after the published samples end. Panel (a): managed factor, net of costs. Panel (b): real-time combination, net of costs.

Table 2 and Figure 2 split the sample at January 2017. Before the split, managed momentum’s gain is large (0.48 to 1.00;  $p = 0.019$  unadjusted, 0.116 after adjusting for six factors in this shorter window). After it, managed momentum’s Sharpe ratio is 0.23 against the original’s 0.21 ( $p = 0.919$ , unadjusted), with a 95% interval for the gain of  $-0.61$  to  $0.66$ . That interval is wide: a test that the gain changed between the two periods gives  $p = 0.118$ . The decade after publication shows no gain, but it cannot rule out the earlier one.

Value is the other case worth a look. After 2016, its net spanning alpha is  $-1.6\%$  a year ( $t = -2.74$ ): the managed factor lost money a static position in value would not have. The Sharpe test is weaker ( $p = 0.065$  unadjusted, 0.388 Holm-adjusted), and with the split moved to January 2016 the alpha falls to  $-1.0\%$  ( $t = -1.61$ ). We read this as a warning, not a finding.

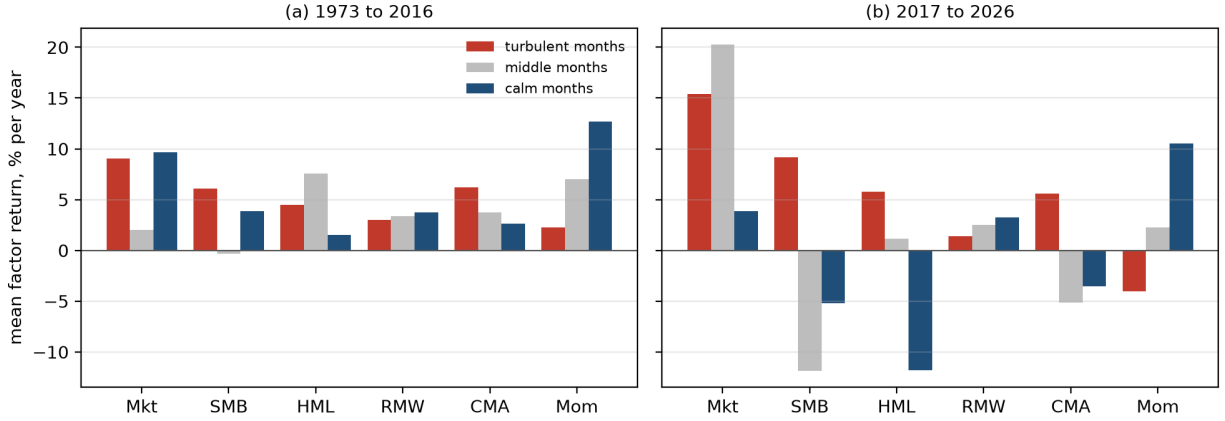
### 3.3 Why: a return channel and a risk channel

| Factor              | Period       | $\bar{w}$ | $\bar{f}$ | $\bar{w}\bar{f}$ | $\widehat{\text{cov}}(w, f)$ | Cost | $\bar{f}^\sigma$ | $M$  | $R$  |
|---------------------|--------------|-----------|-----------|------------------|------------------------------|------|------------------|------|------|
| Market              | 1973 to 2016 | 0.64      | 6.9       | 4.4              | 0.3                          | 0.5  | 4.3              | 0.97 | 1.15 |
|                     | 2017 to 2026 | 0.72      | 13.1      | 9.4              | $-3.1$                       | 0.6  | 5.7              | 0.61 | 1.19 |
| Size (SMB)          | 1973 to 2016 | 0.51      | 3.2       | 1.6              | 0.0                          | 0.4  | 1.3              | 0.77 | 1.06 |
|                     | 2017 to 2026 | 0.36      | $-2.6$    | $-0.9$           | $-1.2$                       | 0.3  | $-2.4$           | n/a  | 0.80 |
| Value (HML)         | 1973 to 2016 | 0.75      | 4.5       | 3.4              | $-0.8$                       | 0.5  | 2.1              | 0.63 | 1.17 |
|                     | 2017 to 2026 | 0.25      | $-1.7$    | $-0.4$           | $-1.2$                       | 0.2  | $-1.9$           | n/a  | 1.09 |
| Profitability (RMW) | 1973 to 2016 | 0.76      | 3.4       | 2.6              | 0.0                          | 0.4  | 2.1              | 0.82 | 1.49 |
|                     | 2017 to 2026 | 0.35      | 2.4       | 0.8              | 0.1                          | 0.3  | 0.7              | 0.80 | 1.28 |
| Investment (CMA)    | 1973 to 2016 | 0.86      | 4.2       | 3.6              | $-0.7$                       | 0.5  | 2.4              | 0.67 | 1.23 |
|                     | 2017 to 2026 | 0.47      | $-1.0$    | $-0.5$           | $-1.2$                       | 0.4  | $-2.0$           | n/a  | 1.03 |
| Momentum            | 1973 to 2016 | 0.76      | 7.3       | 5.6              | 2.4                          | 0.5  | 7.5              | 1.35 | 1.56 |
|                     | 2017 to 2026 | 0.31      | 2.9       | 0.9              | 0.4                          | 0.3  | 0.9              | 1.05 | 1.07 |

**Table 3:** Exact decompositions, equations (6) and (7).  $\bar{w}$  is the average exposure.  $\bar{f}$ ,  $\bar{w}\bar{f}$ , the covariance, the cost and  $\bar{f}^\sigma$  are annualized, in percent, and  $\bar{w}\bar{f} + \widehat{\text{cov}} - \text{cost} = \bar{f}^\sigma$  up to rounding.  $M$  and  $R$  are the return and risk channels; their product is the ratio of the managed to the original Sharpe ratio.  $M$  is not reported when  $\bar{f} \leq 0$ .

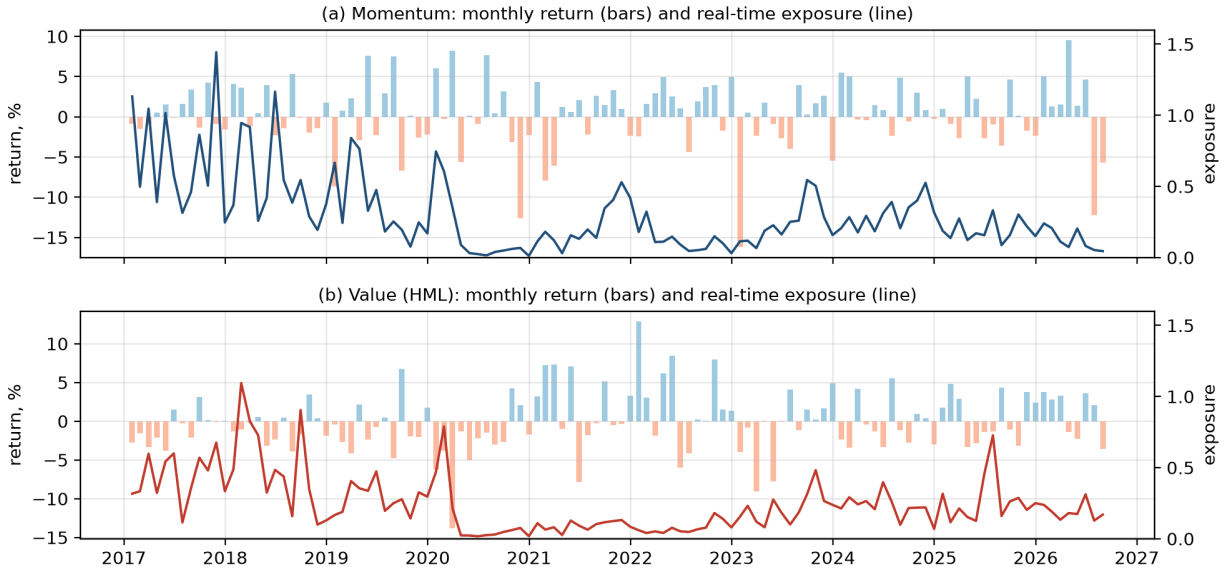
Table 3 applies equations (6) and (7). Before 2017, both channels lift momentum: the covariance adds 2.4 percentage points a year against 0.5 of costs ( $M = 1.35$ ), and cutting exposure in volatile months removes about a third of the risk of a constant exposure ( $R = 1.56$ ). Their product is the doubling of the Sharpe ratio in Table 2. After 2016, both channels are close to one ( $M = 1.05$ ,  $R = 1.07$ ). Momentum’s own return fell from 7.3% to 2.9% a year and the covariance shrank to 0.4 points, most of which the 0.3 points of costs absorbed. And because momentum’s variance stayed high relative to its history, exposure was already low in most months (average 0.31, below 0.2 in 54 of 116 months), so stepping aside in the most volatile ones removed little risk relative to a constant exposure. Value’s covariance is negative in both periods ( $-0.8$  and  $-1.2$  points), so the rule works against it. Profitability shows why a significant spanning alpha need not become a real-time gain: before 2017 its return channel ( $M = 0.82$ ) gives back part of what its risk channel adds ( $R = 1.49$ ), leaving a gain too small to be statistically distinguishable from zero.

Figure 3 shows the covariance directly. Momentum earns most in calm months: after 2016,



**Figure 3:** Annualized mean factor return by tercile of real-time managed exposure. Turbulent months are the third with the lowest exposure, that is, the highest prior-month variance.

10.5% a year in the calm tercile against  $-4.0\%$  in the turbulent one. Value is the mirror image, earning 5.8% in turbulent months and  $-11.8\%$  in calm ones. Figure 4 shows the consequence for value: its eight best months since 2017, 7 of them in the 2021 to 2022 rebound, came while the managed strategy held between 0.04 and 0.28 times the factor.



**Figure 4:** Monthly factor return (bars) and real-time managed exposure (line, cap 1.5), January 2017 to August 2026.

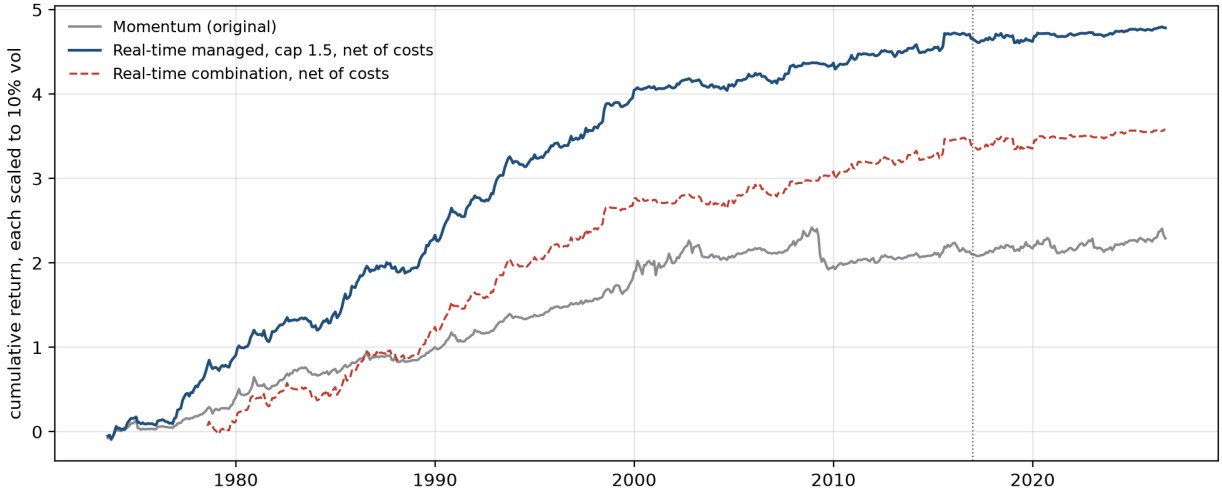
### 3.4 Robustness

Table 4 varies the bootstrap block length, the split date, the cap, the scaling rule and the start date. The burn-in needs no separate check: the scale constant uses every month since 1963 whatever its length, so it only decides when trading starts. Managed momentum's full-sample Sharpe ratio

| Specification             | Momentum, full sample |         |       | Value, after the split |       |       |
|---------------------------|-----------------------|---------|-------|------------------------|-------|-------|
|                           | Original              | Managed | $p$   | $\alpha$ , %           | $t$   | $p$   |
| Baseline                  | 0.43                  | 0.90    | 0.005 | -1.6                   | -2.74 | 0.065 |
| 6-month bootstrap blocks  | 0.43                  | 0.90    | 0.004 | -1.6                   | -2.74 | 0.044 |
| 24-month bootstrap blocks | 0.43                  | 0.90    | 0.005 | -1.6                   | -2.74 | 0.092 |
| Split at January 2016     | 0.43                  | 0.90    | 0.005 | -1.0                   | -1.61 | 0.216 |
| Cap 2                     | 0.43                  | 0.92    | 0.005 | -1.6                   | -2.74 | 0.065 |
| No cap                    | 0.43                  | 0.89    | 0.006 | -1.6                   | -2.74 | 0.065 |
| Inverse volatility        | 0.43                  | 0.76    | 0.013 | -1.7                   | -2.14 | 0.097 |
| Start in July 1978        | 0.42                  | 0.85    | 0.013 | -1.6                   | -2.74 | 0.065 |

**Table 4:** Robustness. Momentum columns test the managed against the original factor over the full real-time sample (the last row starts in July 1978); value columns report the net spanning alpha and the Sharpe test after the split (January 2017 unless stated).  $p$ -values are unadjusted. The value alpha does not change across caps because value’s exposure never exceeds 1.5 after 2016.

ranges from 0.76 to 0.92 and every test rejects equality ( $p \leq 0.013$ ). The value result depends on the split date, which is why we do not rely on it. Figure 5 shows momentum’s path at equal volatility.



**Figure 5:** Cumulative summed monthly returns of momentum, its real-time managed version and the real-time combination (both net of costs), each scaled to 10% annualized volatility for display. Scaling leaves every Sharpe ratio unchanged. The dotted line marks January 2017.

## 4 Conclusion

Built with only past data, capped leverage and costs, volatility management improves one of the six most used equity factors, momentum, and that improvement is concentrated in the years before the rule was published. In the decade since, managed momentum has matched the original. Over the full sample, no real-time combination beat its original factor after costs. Equation (7) gives investors a check they can run before adopting the rule: it pays only when the product of the two channels exceeds one, that is, when the return earned in the months the rule holds more of a factor,

net of costs, and the risk removed by stepping aside in volatile months together outweigh a constant exposure.

Two limits apply. The post-2016 window has 116 months, enough to detect a large change but not a moderate one. We charge costs only on changes in exposure, not on the trading inside each factor portfolio, so every net figure is an upper bound.

**Data availability.** Factor returns are publicly available from the Kenneth R. French Data Library. Code is available from the author on request.

**Declaration of generative AI and AI-assisted technologies in the writing process.** During the preparation of this work the author used generative AI tools to create the simulations and code and to draft the content structure of the paper. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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